

Lab 10 Pair Programming – Graphics with JavaFX

Aim

This week lab is focused on applying OOP techniques in drawing shapes with JavaFX.

One effective method for cultivating computational thinking skills while learning programming in computer labs is through Pair Programming. In this approach, two students collaborate on solving the same problem. One student assumes the role of the Driver, responsible for writing the code and translating design concepts into actual code implementation. Meanwhile, the other student, known as the Observer, reviews the code, identifies errors, and provides design suggestions and improvements. The roles of the driver and observer are continuously switched throughout the session.

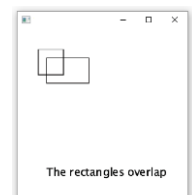
在计算机实验室学习编程时，培养计算思维技能的一种有效方法是采用“结对编程”。在这种方法中，两名学生合作解决同一个问题。一名学生扮演“驾驶员”的角色，负责编写代码并将设计概念转化为实际的代码实现。与此同时，另一名学生，被称为“观察员”，他审查代码，识别错误，并提供设计、改进建议。在整个会话中，“驾驶员”和“观察员”的角色不断交替。

Pair Programming Lab:

1. The Lab TA will create a pair within the group.
2. Talk to your pair, who is going to be the Driver first.
3. Complete Task 1 with Pair Programming (30-45 minutes).
4. Show the TA the results of your works in Task 1.
5. Switch roles.
The Observer becomes the Driver and vice versa.
6. Complete Task 2 with Pair Programming (30-45 minutes).
7. Report to TA after you have completed Task 2.
8. TA gives grade.

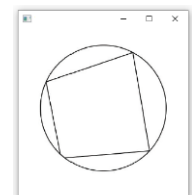
Pair Programming Lab 10 Task 1 - Draw overlapping rectangles + text

- Write a program that displays an **Overlapping Rectangle + Text** with JavaFX.
- The rectangle and text should grow/shrink as the window is resized.



Pair Programming Lab 10 Task 2 - Draw a polygon inside a circle

- Switch Driver $\leftrightarrow$ Observer.
- Write a program that displays **Any Polygon inside a Circle** with JavaFX.
- The polygon inside a circle should grow/shrink as the window is resized.



Optional: If you still have time, draw a Smiley Face that can be resized!

This is the end of CPT111-2324 Pair Programming Lab 10 Task Sheet.